Q1 A 20-hp 240-V 76-A 900 r/min series motor has a field winding of 33 turns per pole. Its armature resistance is 0.09 Ω, and its field resistance is 0.06 Ω. The magnetization curve expressed in terms of magnetomotive force versus E_A at 900 r/min is given by the following table:

E_A, V	95	150	188	212	229	243
$\mathcal{F}$, A · turns	500	1000	1500	2000	2500	3000

Armature reaction is negligible in this machine.

(a) Compute the motor's torque, speed, and output power at 33, 67, 100, and 133 percent of full-load armature current. (Neglect rotational losses.)

(b) Plot the terminal characteristic of this machine.

Q2 A 300-hp 440-V 560-A, 863 r/min shunt dc motor has been tested, and the following data were taken:

Blocked-rotor test:

V_A = 16 3 . V exclusive of brushes V_F = 440 V

I_A = 500 A I_F = 8.86 A

No-load operation:

V_A = 16 3 . V including brushes I_F = 8.76 A

I_A = 23.1 A n = 863 r/min

What is this motor's efficiency at the rated conditions? [*Note*: Assume that (1) the brush voltage drop is 2 V; (2) the core loss is to be determined at an armature voltage equal to the armature voltage under full load; and (3) stray load losses are 1 percent of full load.]

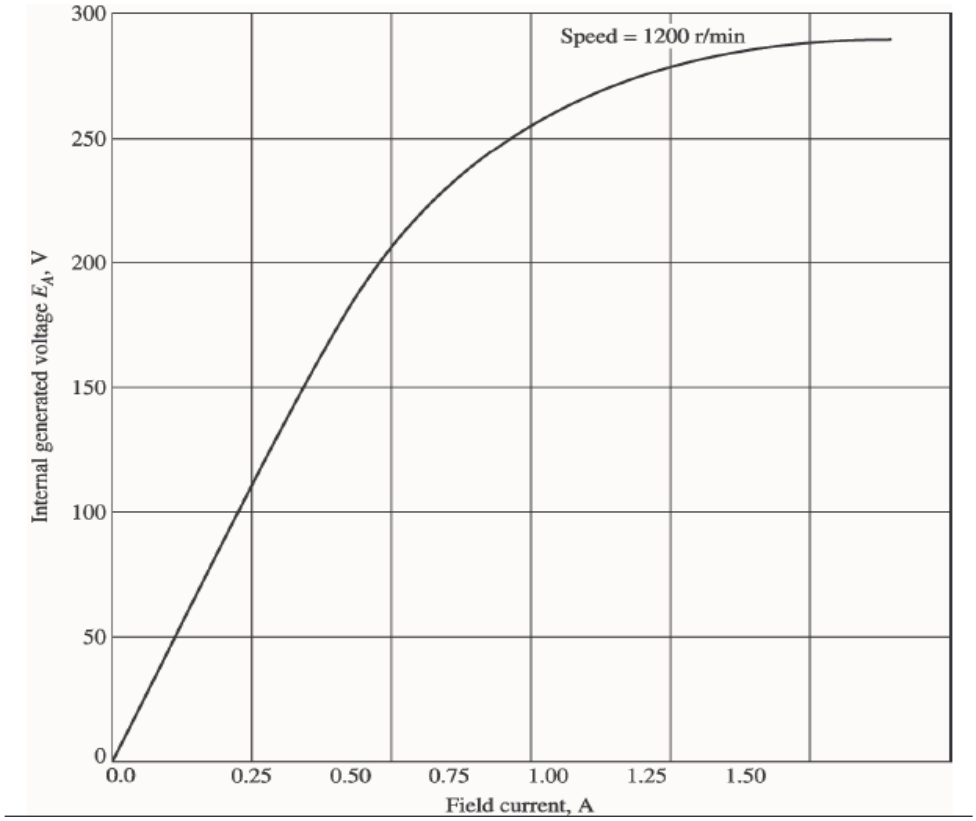
Q3 A 240-V 100-A dc Shunt motor has following parameters

R_A = 0.14 Ω N_F = 1500 turns

R_F = 200 Ω n_m = 1200 r/min

R_{adj} = 0 to 300 Ω, currently set to 120 Ω

This motor has compensating windings and interpoles. The magnetization curve for this motor at 1200 r/min is shown in Figure below



(a) What is the no-load speed of this motor when *R*adj = 120 Ω?

(b) What is its full-load speed?

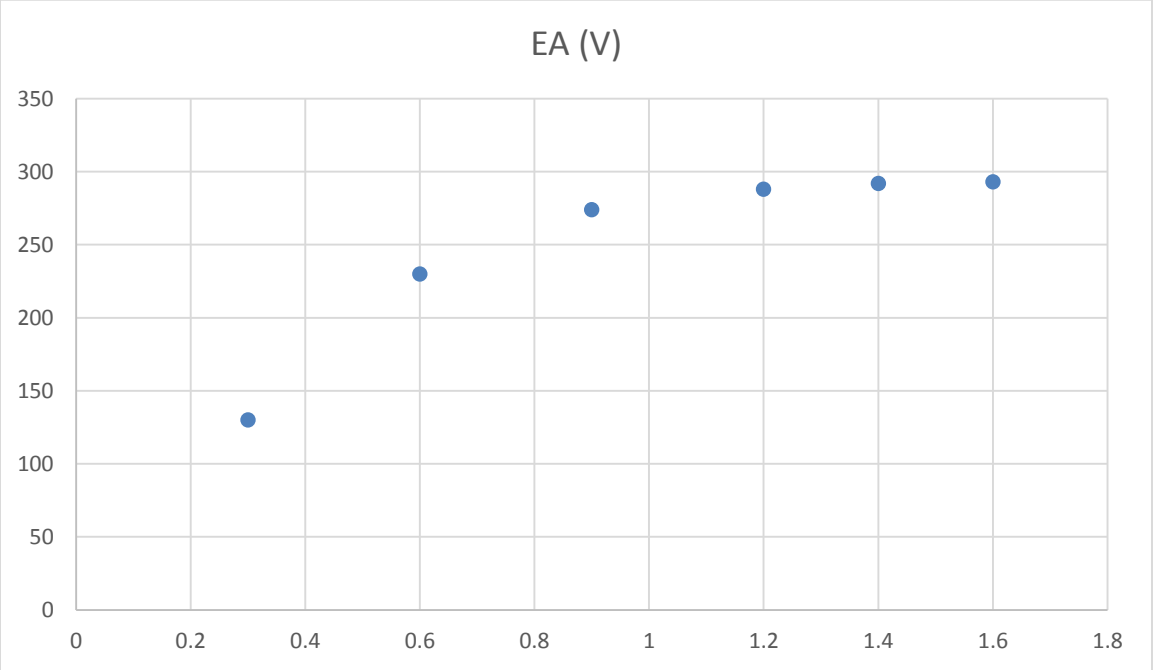
(c) Under no-load conditions, what range of possible speeds can be achieved by adjusting *R*adj

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Q4) A 15KW, 240V compensated DC shunt generator has armature resistance of 0.45 Ω. Its field resistance is 100 Ω and adjustable resistance connected in series with field coil can vary from 50 Ω to 700 Ω.
Its no load characteristic curve is shown below with If(in A) at x axis and EA(in volts) at y axis.

Find out following quantities assuming rated terminal voltage in all cases.

a) Internally generated voltage at a field current of 1 A
b) Value of adjustable resistance connected in series with field coil
c) Terminal voltage when load current is 45A.
d) Input power of the generator if the core losses are 400W.



Q5) Consider a 250-V 120-A dc Shunt motor having following parameters

R_A = 0.18 Ω N_F = 1400 turns N_A = 90 turns

R_F = 200 Ω n_m = 1250 r/min

R_{adj} = 10 to 400 Ω,

Find out following

1. Range of possible speeds at no load by adjusting R_{adj}
2. What is full-load speed of this motor when R_{adj} = 100 Ω, 200 Ω, 300 Ω respectively?
3. What is induced torque in above part?
4. What is the output power and load torque if core losses=500watts, mechanical losses=400watts

This motor has compensating windings and interpoles. The magnetization curve for this motor is shown in Figure below

Magnetization Curve for above motor at speed = 1200 rpm

I_F (A)	0.25	0.5	0.75	1	1.25
E_A (V)	100	200	270	285	290

Solve above mentioned problems plus end problems 8-1 to 8-12 from the text book as assignment2.